

Numerical study of the interface errors of finite-difference simulations of seismic waves

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ABSTRACT

Numerical simulations of wave propagation produce different errors and the most well known is numerical dispersion, which is only valid for homogeneous media. However, there is a lack of error studies for heterogeneous media or even for the canonical case of media that have two constant velocity layers. The error associated with media that have two layers is called an interface error, and it typically converges to zero with a lower order of convergence compared to the theoretical convergence rate of the finite-difference schemes (FDS) for homogeneous media. We evaluated a detailed numerical study of the interface error for three staggered-grid FDS that are commonly used in the simulation of seismic-wave propagation.

We determined that a standard staggered-grid scheme (SSGS) (also known as the Virieux scheme), a rotated staggered-grid scheme (RSGS), and a Lebedev scheme (LS) preserve the second order of convergence at horizontal/vertical solid-solid interfaces when the medium parameters have been properly modified, such as by harmonic averaging of finely layered media for the stiffness tensor and arithmetic mean for the density. However, for a fluid-solid interface aligned with the grid line, a second-order convergence can only be achieved by an SSGS. In addition, the presence of a fluid-solid interface reduces the order of convergence for the LS and the RSGS to a first order of convergence. The presence of inclined interfaces makes high-order (second and more) convergence impossible.

INTRODUCTION

Currently, full-waveform numerical simulations are one of the main tools for studying seismic-wave propagation in 3D realistic inhomogeneous isotropic/anisotropic elastic media. There are several principal numerical methods used to compute a wavefield such as finite differences (FDs) (Lebedev, 1964; Alford et al., 1974; Kelly et al., 1976; Virieux, 1986; Levander, 1988; Saenger et al., 2000), finite elements (FEs) (Zhang and Verschuur, 2002), spectral elements (SE) (Komatitsch and Vilotte, 1998; Tromp et al., 2008), discontinuous Galerkin (DG) (Grote et al., 2006; Etienne et al., 2010), finite volumes (FVs) (Zhang, 1997), and combinations of these methods (Dumbser and Kaser, 2006; Kaser and Dumbser,

2006; Zhang et al., 2012). Galerkin-type methods (FE, DG, and SE), in principle, can provide an arbitrarily high accuracy by increasing the dimension of the functional space used. Moreover, they can be applied on irregular grids, thus allowing the handling of interfaces with almost arbitrary geometries. However, to achieve the desired accuracy, the grids must precisely match at least the main interfaces. Grid generation is difficult to automate and is time consuming. On the contrary, FD schemes (FDS) are a universal method that can be applied to regular rectangular grids, which makes them attractive for numerical simulations of seismic-wave propagation in realistic 3D heterogeneous media and is the reason that FD methods are being intensively developed and applied. FDS can be roughly divided into the following types:

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- 1) conventional central difference schemes for second-order elastic-wave equations (displacement formulation) (Alford et al., 1974; Kelly et al., 1976);
- 2) staggered-grid schemes for first-order formulation of elastic-wave equation in velocity/displacement–stress formulations (Virieux, 1986; Levander, 1988);
- 3) rotated staggered-grid and Lebedev schemes (LSs) for elastic-wave equation in velocity/displacement–stress formulation for anisotropic media (Saenger et al., 2000; Asvadurov et al., 2007). A detailed review and comparative analysis of the main classes of FDS can be found in Moczo et al. (2010, 2011), and Virieux et al. (2011).

Numerical solutions to the wave equation produce different types of errors. Initially, attention is directed at the numerical dispersion caused by the frequency dependence of wave propagation velocity for an FD/FE approximation of an elastic-wave equation. This dispersion is indicated by the distortion of a propagating plane-wave pulse and has been thoroughly studied. There are several approaches for its reduction, such as by increasing the order of FD approximations of the differential operators (Levander, 1988) and using the Lax-Wendroff correction (Dablain, 1986) and direct dispersion reduction (Min et al., 2000).

In this paper, we do not consider dispersion analyses; instead, we focus on the errors introduced by FD approximations of elastic-wave equations within media that have interfaces. These errors are not as noticeable as numerical dispersions because they do not distort the shape of the wave, despite producing incorrect reflection/transmission coefficients and critical angles that generate incorrect head waves.

There are two methods of introducing interfaces when numerically modeling wave propagation. The first method is based on describing an interface and approximating the corresponding contact conditions. FDS based on this approach are known as *homogeneous*. However, this method is problematic because the number of interfaces can be large and the interfaces can have complicated intersecting shapes. Moreover, FDS require the use of regular meshes or those that can be mapped onto a regular one; thus, such techniques are applied now to approximate specific interfaces such as the free surface (Hestholm and Ruud, 2000, 2002; Tarras et al., 2011). Therefore, the most widespread method of introducing interfaces in practice is the use of *heterogeneous* FDS that apply the same formulas everywhere on the grid and have modified coefficients in the vicinity of the interface. This modification is performed to provide the desired rate of convergence of the FD solution to the exact solution. The analysis of different approaches to modifying FDS based on the second-order elastic-wave equation was thoroughly discussed in the 1980s and 1990s by Stephen (1983), Kummer et al. (1987), Zahradnik et al. (1993), and others. The same analysis for a staggered-grid-based FDS approximating the velocity-stress formulation of the isotropic-wave equation (Virieux and Madariaga, 1982; Virieux, 1986) can be found in Muir et al. (1992), Cohen and Joly (1996), Stephen (1996), Moczo et al. (2002), Gustafsson and Mossberg (2004), and Gustafsson and Wahlund (2004). Moczo et al. (2002) present two methods of treating the interface error of the standard staggered-grid scheme (SSGS) (Virieux, 1986). When a solid-solid interface is aligned to the grid line, the arithmetic mean for the density and the harmonic mean of the shear modulus were derived according to the integral conservation law. This approach is known as a *balance technique* and is

presented in Kummer et al. (1987) and Samarskii (2001). For the Lebedev FDS, the approaches are similar to those applied for averaging finely layered media (Backus, 1962; Schoenberg and Muir, 1989). A discussion of the accuracy and grid treatment of the staggered-grid scheme for a rough and fluid-free surface (waves on the ocean) is given in Stephen (1983).

In recent years, interest in how the interface affects the accuracy of FDS has increased and began with analysis published in papers by Symes (2007) and Symes and Vdovina (2009) and is further developed by Lisitsa et al. (2010), Podgornova and Lisitsa (2010), and Podgornova et al. (2010). In particular, Symes (2007) and Symes and Vdovina (2009) conclude that FDS based on staggered grid possess an interface error of the first order. Later, Lisitsa et al. (2010) show that there is a unique way to modify the coefficients of the scheme in the vicinity of the interface providing a second-order decay of the interface error with grid refinement. The results presented in Lisitsa et al. (2010) are derived for a 1D problem, and as expected, they are consistent with the results of Samarskii (2001) and Moczo et al. (2002).

However, the interface error of staggered-grid schemes has only been studied for contacts of two isotropic media (Virieux, 1986), and simulations of seismic-wave propagation in anisotropic media are getting to be of high importance for practice. Therefore, an analysis is needed for two FDS most commonly used for the approximation of the anisotropic-wave equation: the rotated staggered-grid scheme (RSGS) (Saenger et al., 2000) and the LS (Lebedev, 1964; Davydycheva et al., 2003; Asvadurov et al., 2007). The main advantage of the RSGS is that the stress tensor is defined in the center of the grid cell, and no modification of the stiffness tensor is required when an interface is aligned to the grid. However, following Lisitsa and Vishnevskiy (2010) and Bernth and Chapman (2011), it can be concluded that the LS is better suited for large-scale numerical simulations of seismic waves in anisotropic media because it consumes less memory. However, the geometry of the grid cell for the LS requires a modification of the stiffness tensor components. According to the results presented in Podgornova et al., (2010), to preserve the second-order convergence, the formulas for averaging finely layered media (Schoenberg and Muir, 1989) must be applied to modify the model parameters in the vicinity of the interfaces that are parallel to the grid lines. This study is done by *reflectivity analysis*, which was based on the direct comparison of plane-wave reflection/transmission coefficients for plane interfaces in differential and FD formulations.

This paper is focused on the interface error analysis for heterogeneous FDS of seismic-wave simulation in anisotropic media, and it begins with an overview of the main approaches to modifying the coefficients of FDS: the balance technique, averaging of finely layered media, and reflectivity analysis (a description of these approaches is presented in Appendix A). Next, we apply these methods to the SSGS, LS, and RSGS and verify the rate of interior convergence of FD solutions (the notion of interior convergence is explained in Appendix B) by a series of numerical experiments for horizontal/vertical interfaces between two isotropic/anisotropic solids. The results of numerical experiments for the same FDS without modified coefficients are presented to show that these experiments produce first-order interior convergence, and we also present snapshots of fluid-isotropic solid interfaces that demonstrate the advantages of proper coefficient modification.

All of the approaches to modifying FDS coefficients in the presence of interfaces are rigorously substantiated for the planar interfaces aligned to the grid lines. Inclined planar interfaces are treated as staircase lines/surfaces, so the accuracy of their geometrical approximation is of the first order; therefore, the convergence will not be higher than the first order, which is clearly reflected by the numerical experiments presented in this paper.

PRELIMINARIES

Consider the velocity-stress formulation of the elastic-wave equation for the 2D Cartesian coordinates:

$$\begin{aligned}\rho \frac{\partial u_x}{\partial t} &= \frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xz}}{\partial z}, \\ \rho \frac{\partial u_z}{\partial t} &= \frac{\partial \sigma_{xz}}{\partial x} + \frac{\partial \sigma_{zz}}{\partial z},\end{aligned}\quad (1)$$

$$\begin{pmatrix} \frac{\partial \sigma_{xx}}{\partial t} \\ \frac{\partial \sigma_{xz}}{\partial t} \\ \frac{\partial \sigma_{zz}}{\partial t} \end{pmatrix} = \begin{pmatrix} c_{11} & c_{12} & c_{13} \\ c_{12} & c_{22} & c_{23} \\ c_{13} & c_{23} & c_{33} \end{pmatrix} \begin{pmatrix} \frac{\partial u_x}{\partial x} \\ \frac{\partial u_z}{\partial z} \\ \frac{\partial u_z}{\partial x} + \frac{\partial u_x}{\partial z} \end{pmatrix}. \quad (2)$$

In these notations, u_x and u_z are components of the velocity vector; σ_{xx} , etc. are components of the stress tensor; ρ is the mass density; and $C = c_{ij}$ is the stiffness tensor, written in matrix form. If $c_{13} = c_{23} = 0$, then the stiffness tensor has a block-diagonal structure that is used for the construction of the SSGS. In the 2D equation, this corresponds to isotropic and orthotropic models. In the 3D statement, it corresponds to isotropic, transversally isotropic, and orthorhombic models.

The main goal of this paper is to analyze interior convergence of certain FD approximations for a continuous elastic medium with interfaces S_j where conditions of an ideal contact are valid:

$$[u_x]|_{S_j} = [u_z]|_{S_j} = 0, \quad (3)$$

$$[\sigma_{xx}n_x + \sigma_{xz}n_z]|_{S_j} = [\sigma_{xz}n_x + \sigma_{zz}n_z]|_{S_j} = 0, \quad (4)$$

where $[\phi]|_{S_j}$ is the jump of ϕ across S_j .

Finite-difference approximations

Here, we provide a brief description of the three explicit time-domain staggered-grid FDS widely used to approximate the elastic-wave equation in velocity-stress formulations: an SSGS or the Virieux scheme (Virieux, 1986; Levander, 1988), an RSGS (Saenger et al., 2000), and the LS (Lebedev, 1964; Asvadurov et al., 2007). The first is designed for isotropic elastic media, whereas the second and the third are focused on anisotropic media.

Before discussing the details of a particular scheme, it is important to introduce a grid with integer points $x_i = ih_x$, $z_j = jh_z$, and $t^n = n\tau$ and half-integer points $x_{i+1/2} = (i + 1/2)h_x$, $z_{j+1/2} = (j + 1/2)h_z$, and $t^{n+1/2} = (n + 1/2)\tau$, where τ , h_x , and h_z are the grid steps with respect to time and space. Hereafter, the lower-case indices are integers and the uppercase indices are either

integers or half-integers. We will use the following notation for the grid functions: $f_{I,J}^N = f(t^N, x_I, z_J)$.

It is also convenient to define the FD operators that are used to construct the above-mentioned schemes:

$$D_t[f]_{I,J}^N = \frac{f_{I,J}^{N+1/2} - f_{I,J}^{N-1/2}}{\tau} = \frac{\partial f}{\partial t}|_{x_I, y_J}^{t_N} + O(\tau^2), \quad (5)$$

$$\begin{aligned}D_x^L[f]_{I,J}^N &= \sum_{m=0}^M \alpha_m \frac{f_{I+m+1/2,J}^N - f_{I-m-1/2,J}^N}{h_x} \\ &= \frac{\partial f}{\partial x}|_{x_I, y_J}^{t_N} + O(h_x^q),\end{aligned}\quad (6)$$

$$\begin{aligned}D_x^R[f]_{I,J}^N &= \sum_{m=0}^M \frac{\alpha_m}{2} \frac{f_{I+m+1/2,J+m+1/2}^N - f_{I-m-1/2,J-m-1/2}^N}{h_x} \\ &\quad + \sum_{m=0}^M \frac{\alpha_m}{2} \frac{f_{I+m+1/2,J-m-1/2}^N - f_{I-m-1/2,J+m+1/2}^N}{h_x} \\ &= \frac{\partial f}{\partial x}|_{x_I, y_J}^{t_N} + O(h_x^q).\end{aligned}\quad (7)$$

Operators approximating the derivatives with respect to z can be constructed from relations 6 and 7. Parameter q in formulas 6 and 7 depends on the choice of the coefficients α_m , and their number can vary from 2 up to $2(M+1)$ if, for example, the coefficients were derived to minimize the numerical dispersion for coarse discretizations (Min et al., 2000).

Standard staggered-grid scheme

An SSGS, or the Virieux scheme, is introduced in Virieux (1986) for the numerical simulation of waves in isotropic elastic media. The essence of this approach consists of assigning different components of the numerical solution (velocities and stresses) to different points with respect to time and space. The grid for the SSGS is presented in Figure 1. As a result, the SSGS can be constructed using operators 5 and 6:

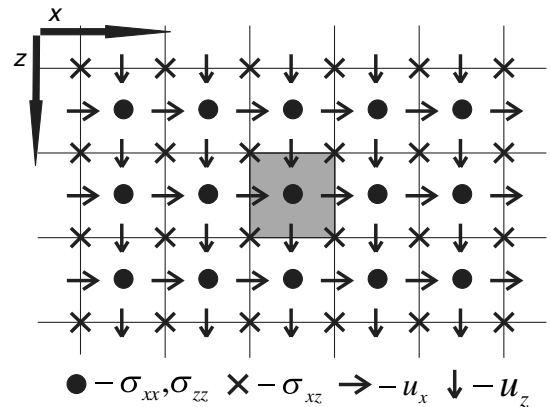


Figure 1. The grid for the SSGS. Components σ_{xx} , σ_{zz} , and the model parameters are placed in the circles, σ_{xz} corresponds to crosses, and horizontal/vertical velocities are defined at horizontal/vertical arrows. The grid cell is represented by the gray square.

$$\begin{aligned}
(\hat{\rho})_{i+1/2,j} D_t [u_x]_{i+1/2,j}^{n-1/2} &= D_x^L [\sigma_{xx}]_{i+1/2,j}^{n-1/2} + D_z^L [\sigma_{xz}]_{i+1/2,j}^{n-1/2}, \\
(\hat{\rho})_{i,j+1/2} D_t [u_z]_{i,j+1/2}^{n-1/2} &= D_x^L [\sigma_{xz}]_{i,j+1/2}^{n-1/2} + D_z^L [\sigma_{zz}]_{i,j+1/2}^{n-1/2}, \\
D_t [\sigma_{xx}]_{i,j}^n &= (\hat{c}_{11})_{i,j} D_x^L [u_x]_{i,j}^n + (\hat{c}_{12})_{i,j} D_z^L [u_z]_{i,j}^n, \\
D_t [\sigma_{zz}]_{i,j}^n &= (\hat{c}_{12})_{i,j} D_x^L [u_x]_{i,j}^n + (\hat{c}_{22})_{i,j} D_z^L [u_z]_{i,j}^n, \\
D_t [\sigma_{xz}]_{i+1/2,j+1/2}^n &= (\hat{c}_{33})_{i+1/2,j+1/2} (D_x^L [u_z]_{i+1/2,j+1/2}^n + D_z^L [u_x]_{i+1/2,j+1/2}^n), \quad (8)
\end{aligned}$$

where n , i , and j are integers. Hats are placed over the model parameters to indicate that values such as $\hat{\rho}_{i,j+1/2}$ are not the values of these functions at the corresponding points but values that will be specified below.

Lebedev scheme

To construct the LS approximating the elastic-wave equations 1 and 2, two sets of points must be introduced:

- 1) a set with an integer sum of spatial indices, which is $\Omega_\sigma^L = \{(x_i, z_j)\} \cup \{(x_{i+1/2}, z_{j+1/2})\}$, and
- 2) a set with a half-integer sum of indices, which is $\Omega_u^L = \{(x_{i+1/2}, z_j)\} \cup \{(x_i, z_{j+1/2})\}$. The grid for the LS is shown in Figure 2. Next, defining all stresses in Ω_σ^L and all velocities in Ω_u^L , one may write the LS with the help of operators 5 and 6 as follows:

$$\begin{aligned}
\hat{\rho}_{I,J} D_t [u_x]_{I,J}^{n-1/2} &= D_x^L [\sigma_{xx}]_{I,J}^{n-1/2} + D_z^L [\sigma_{xz}]_{I,J}^{n-1/2}, \quad (x_I, z_J) \in \Omega_u^L, \\
\hat{\rho}_{I,J} D_t [u_z]_{I,J}^{n-1/2} &= D_x^L [\sigma_{xz}]_{I,J}^{n-1/2} + D_z^L [\sigma_{zz}]_{I,J}^{n-1/2}, \quad (x_I, z_J) \in \Omega_u^L, \\
\begin{pmatrix} D_t [\sigma_{xx}]_{I,J}^n \\ D_t [\sigma_{zz}]_{I,J}^n \\ D_t [\sigma_{xz}]_{I,J}^n \end{pmatrix} &= \hat{C}_{I,J} \begin{pmatrix} D_x^L [u_x]_{I,J}^n \\ D_z^L [u_z]_{I,J}^n \\ D_x^L [u_z]_{I,J}^n + D_z^L [u_x]_{I,J}^n \end{pmatrix}, \quad (x_I, z_J) \in \Omega_\sigma^L, \quad (9)
\end{aligned}$$

where I and J can be either integers or half-integers, but n is an integer. Parameter $\hat{\rho}_{I,J}$ and tensor $\hat{C}_{I,J}$ will be introduced later.

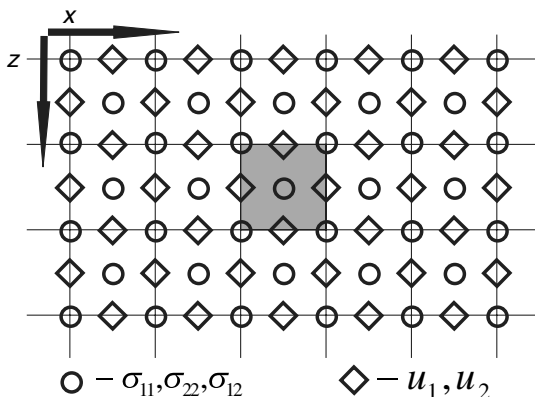


Figure 2. The grid for the LS. All components of the stress tensor are placed in the circles, and components of the velocity vector are defined in diamonds. The grid cell is represented by the gray square.

Rotated staggered-grid scheme

To construct the rotated staggered-grid scheme (RSGS), two sets of points must be considered:

- 1) a set with integer indices, which is $\Omega_\sigma^R = \{(x_i, z_j)\}$, and
- 2) a set with fractional indices, which is $\Omega_u^R = \{(x_{i+1/2}, z_{j+1/2})\}$. All components of the stress tensor must be defined in Ω_σ^R , whereas the velocities must be defined in Ω_u^R . The grid for the RSGS is presented in Figure 3. Applying operators 5 and 7 results in the following:

$$\begin{aligned}
\hat{\rho}_{I,J} D_t [u_x]_{I,J}^{n-1/2} &= D_x^R [\sigma_{xx}]_{I,J}^{n-1/2} + D_z^R [\sigma_{xz}]_{I,J}^{n-1/2}, \quad (x_I, z_J) \in \Omega_u^R, \\
\hat{\rho}_{I,J} D_t [u_z]_{I,J}^{n-1/2} &= D_x^R [\sigma_{xz}]_{I,J}^{n-1/2} + D_z^R [\sigma_{zz}]_{I,J}^{n-1/2}, \quad (x_I, z_J) \in \Omega_u^R, \\
\begin{pmatrix} D_t [\sigma_{xx}]_{I,J}^n \\ D_t [\sigma_{zz}]_{I,J}^n \\ D_t [\sigma_{xz}]_{I,J}^n \end{pmatrix} &= \hat{C}_{I,J} \begin{pmatrix} D_x^R [u_x]_{I,J}^n \\ D_z^R [u_z]_{I,J}^n \\ D_x^R [u_z]_{I,J}^n + D_z^R [u_x]_{I,J}^n \end{pmatrix}, \quad (x_I, z_J) \in \Omega_\sigma^R. \quad (10)
\end{aligned}$$

NUMERICAL EXPERIMENTS

This section presents the results of numerical experiments the compare that rate of convergence of FD solutions to that of differential formulations with the following interfaces:

- 1) a planar interface aligned to grid lines, which subsequently is referred to as the *horizontal interface*;
- 2) an inclined planar interface;
- 3) a contact with three media, which is subsequently referred to as the *corner problem* (Figure 12).

To study the convergence rate, a set of nested grids with the grid steps $h_k = 2^{-k} h_0$ for $h_0 = 1$ m is introduced. The refinement is performed up to $h_7 = h_0/128 \approx 8$ mm. Subsequently a sequence of L_2 relative errors

$$\epsilon_k = \|v_k - v_{k-1}\|_2 / \|v_k\|_2 \quad (11)$$

is computed with v_k as the FDS solution on the k th grid. The parameter

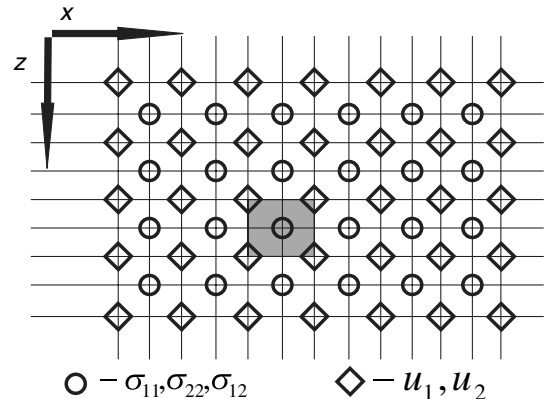


Figure 3. The grid for the RSGS. All components of the stress tensor are placed in the circles, and components of the velocity vector are defined at diamonds. The grid cell is represented by the gray square.

$$\delta_k = \epsilon_k / \epsilon_{k-1} \quad (12)$$

is hereafter referred to as the convergence indicator. For a linear FDS, it can be proven that

$$\delta_k \approx \left(\frac{h_k}{h_{k+1}} \right)^\gamma, \quad (13)$$

where γ is the convergence rate of the numerical solution to the solution of the differential statement (see Appendix B). In our case, $h_k/h_{k+1} = 2$; therefore, if the numerical solution converges with the second-order $\delta_k \approx 4$, the convergence is of the first order if $\delta_k \approx 2$.

For the numerical experiments, the following models were used:

- 1) isotropic solid 1 (IS1): $\rho = 1800 \text{ kg/m}^3$, $V_P = 1900 \text{ m/s}$, $V_S = 1200 \text{ m/s}$;
- 2) isotropic solid 2 (IS2): $\rho = 2200 \text{ kg/m}^3$, $V_P = 2400 \text{ m/s}$, $V_S = 1400 \text{ m/s}$;
- 3) isotropic solid 3 (IS3): $\rho = 1600 \text{ kg/m}^3$, $V_P = 1600 \text{ m/s}$, $V_S = 900 \text{ m/s}$;
- 4) ideal fluid (IF): $\rho = 1000 \text{ kg/m}^3$, $V_P = 1500 \text{ m/s}$, $V_S = 0 \text{ m/s}$;
- 5) anisotropic solid 1 (AS1): $\rho = 1800 \text{ kg/m}^3$ and the stiffness tensor is provided by the formulae 14 below,

$$C = \begin{pmatrix} 3.6 & 1.8 & -0.9 \\ 1.8 & 3.24 & 0 \\ -0.9 & 0 & 2.7 \end{pmatrix} \times 10^9 \frac{\text{kg} \times \text{s}}{\text{m}}; \quad (14)$$

- 6) anisotropic solid 2 (AS2): $\rho = 2200 \text{ kg/m}^3$ and the stiffness tensor is provided by the formulae 15 below,

$$C = \begin{pmatrix} 4.4 & 2.2 & 2.2 \\ 2.2 & 6.6 & 2.2 \\ 2.2 & 2.2 & 4.4 \end{pmatrix} \times 10^9 \frac{\text{kg} \times \text{s}}{\text{m}}; \quad (15)$$

- 7) anisotropic solid 3 (AS3): $\rho = 1600 \text{ kg/m}^3$ and the stiffness tensor is provided by the formulae 16 below,

$$C = \begin{pmatrix} 1.6 & 0.8 & 0.8 \\ 0.8 & 2.4 & 0.8 \\ 0.8 & 0.8 & 1.6 \end{pmatrix} \times 10^9 \frac{\text{kg} \times \text{s}}{\text{m}}. \quad (16)$$

The numerical solution was constructed as an approximation for the elastic-wave equation in the velocity-stress formulation with the initial data provided by the formula 17 below:

$$\begin{aligned} u_x &= 0; & u_z &= 0; \\ \sigma_{xz} &= 0; & \sigma_{xx} &= \sigma_{zz} = \exp\{-0.1((x - x_s)^2 + (z - z_s)^2)\}. \end{aligned} \quad (17)$$

The size of the computational domain was $3000 \times 3000 \text{ m}$, and the source was centered at $x_s = 1500 \text{ m}$ and $z_s = 750 \text{ m}$. Artificial reflectionless boundary conditions were not used to avoid an extra error caused by approximation of this type of boundary conditions.

To modify the coefficients of the FDS, the following approaches were used:

- 1) For solid-solid interface for isotropic media: A balance technique was used for SSGS and RSGS and averaging of finely layered models was used for LS.
- 2) For solid-solid interface for isotropic/anisotropic media: A balance technique was used for RSGS, and an averaging of the finely layered model was used for LS.
- 3) For fluid-solid interfaces for all three FDS: The reflectivity analysis was used.

For additional details on these techniques, see Appendix A and the references therein.

The numerical solutions were computed, and the rate of their interior convergence was estimated by constructing the error ϵ_k (equation 11) and the convergence indicator (equation 12).

Horizontal interface

For the interface at a depth of $z = 1500 \text{ m}$, four situations were analyzed:

- 1) Two isotropic solids, with IS1 at the top and IS2 at the bottom. Simulations were performed by the three FDS. The convergence indicators δ_k for these experiments are presented in Figure 4.
- 2) Contact of fluid (IF) and isotropic solid (IS1). Simulations were performed by the three FDS. The convergence indicators δ_k for these experiments are presented in Figure 5.
- 3) Contact of two anisotropic solids, with AS1 at the top and AS2 at the bottom. Wavefields were simulated by the LS and RSGS. The convergence indicators δ_k for these experiments are presented in Figure 6.
- 4) Contact of fluid (IF) and anisotropic solid (AS1). Wavefields were simulated by the LS and RSGS. The convergence indicators δ_k for these experiments are presented in Figure 7.

The experiments confirm the theoretical predictions: For either the isotropic or anisotropic solid-solid contact, all three schemes possess a second-order convergence if the density is modified by the arithmetic averaging over the adjoining grid cells and the

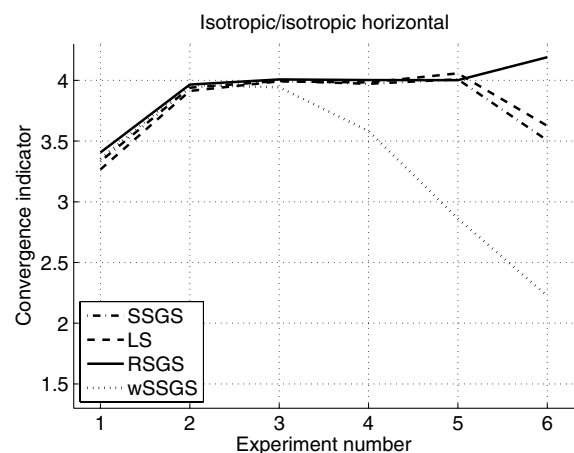


Figure 4. Convergence indicator for the SSGS (dashed-dotted), LS (dashed), RSGS (solid) and SSGS with no parameter modification (wSSGS) (dotted) for the model with a horizontal solid-solid interface, isotropic media. The horizontal axis represents the experiment number, and the vertical axis corresponds to the convergence indicator, which is 4 for the second order and 2 for the first order of convergence.

components of a stiffness tensor are constructed by the formulas for averaging the finely layered media.

For the fluid-solid interface, only the SSGS attains a second-order convergence with $c_{33} = 0$ at the interface, whereas the LS and RSGS produce a first-order convergence.

Inclined interface

The next series of numerical experiments was designed to clarify the convergence with an inclined planar interface for an inclination angle equal to 30° with respect to the horizontal line. We analyzed the same four situations:

- 1) Contact of two isotropic solids IS1 and IS2. The convergence indicator is presented in Figure 8.
- 2) Contact of fluid (IF) and isotropic solid (IS1). The convergence indicator is presented in Figure 9.
- 3) Contact of two anisotropic solids AS1 and AS2. The convergence indicator is presented in Figure 10.
- 4) Contact of fluid (IF) and anisotropic solid (AS1). The convergence indicator is presented in Figure 11.

A staircase approximation of the inclined interface was used, which resulted in the interface being approximated only with the first order. Therefore, the wavefield could not converge faster than

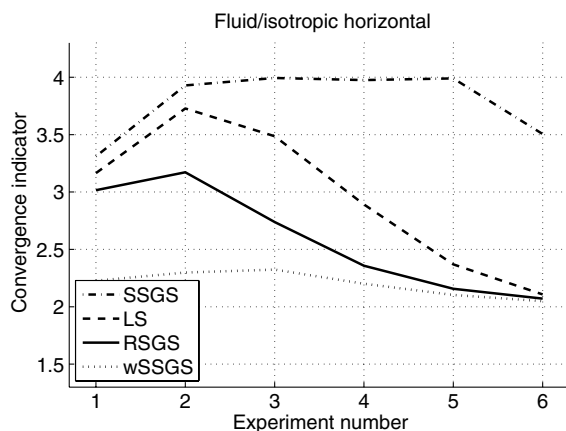


Figure 5. Convergence indicator for the SSGS (dashed-dotted), LS (dashed), RSGS (solid), and SSGS with no parameter modification (wSSGS) (dotted) for the model with a horizontal fluid-solid interface, isotropic media. The horizontal axis represents the experiment number, and the vertical axis corresponds to the convergence indicator, which is 4 for the second order and 2 for the first order of convergence.

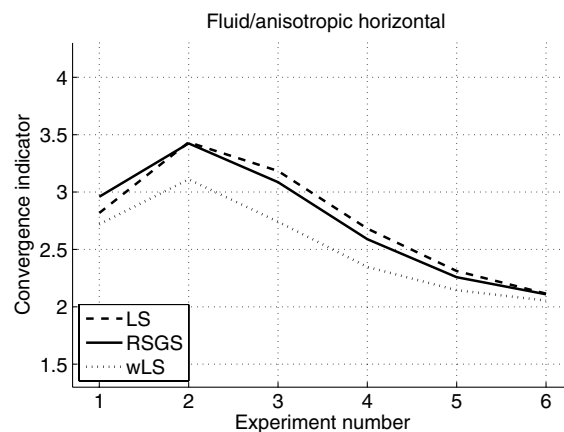


Figure 7. Convergence indicator for the LS (dashed), RSGS (solid), and LS with no parameter modification — WLS (dotted) for the model with a horizontal fluid-solid interface, anisotropic media. The horizontal axis represents the experiment number, and the vertical axis corresponds to the convergence indicator, which is 4 for the second order and 2 for the first order of convergence.

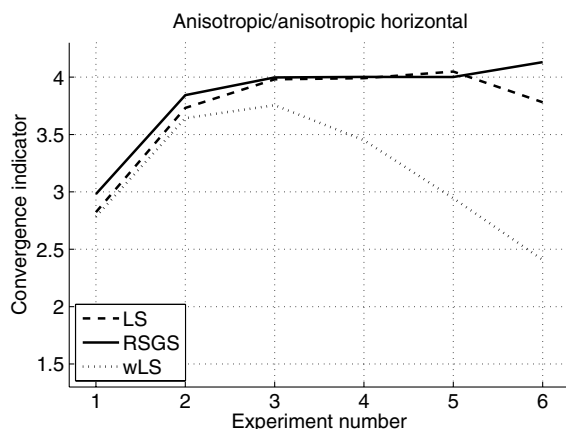


Figure 6. Convergence indicator for the LS (dashed), RSGS (solid), and LS with no parameter modification — WLS (dotted) for the model with a horizontal solid-solid interface, anisotropic media. The horizontal axis represents the experiment number, and the vertical axis corresponds to the convergence indicator, which is 4 for the second order and 2 for the first order of convergence.

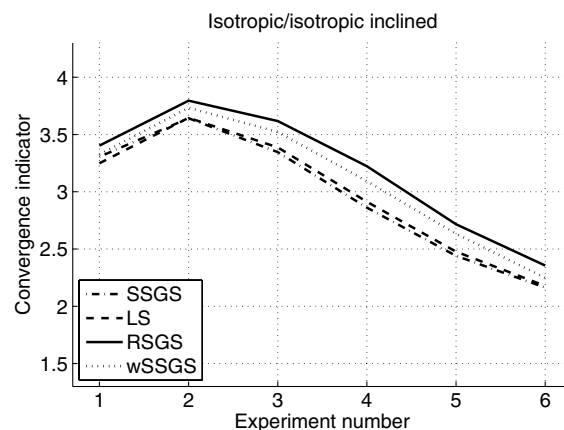


Figure 8. Convergence indicator for the SSGS (dashed-dotted), LS (dashed), RSGS (solid), and SSGS with no parameter modification (wSSGS) (dotted) for the model with an inclined solid-solid interface, isotropic media. The horizontal axis represents the experiment number, and the vertical axis corresponds to the convergence indicator, which is 4 for the second order and 2 for the first order of convergence.

the model, which was confirmed by the numerical experiments. Moreover, we did not even observe a convergence of the first order for the fluid/solid interface, which was likely the result of the staircase approximation of the interface.

Corner problem

The third set of experiments was designed to analyze the typical geometry of subvertical heterogeneities, such as fault structures and sonic-log acquisitions (Pissarenko et al., 2009; Lisitsa and Lys, 2010) (Figure 12). The model is composed of the horizontal layer at $0 < z < 1100$ m, whereas the bottom is divided into two blocks by a vertical interface at $x = 1500$ m (Figure 12). Four experiments were performed in the same manner as the prior experiments:

- 1) Contact of the three isotropic solids, with IS1 at the top, IS2 at the bottom left, and IS3 at the bottom right. The convergence indicator is presented in Figure 13.
- 2) Contact of a fluid and two isotropic solids, with IF at the top, IS1 at the bottom left, and IS2 at the bottom right. The convergence indicator is presented in Figure 14.
- 3) Contact of three anisotropic solids, with AS1 at the top, AS2 at the bottom left, and AS3 at the bottom right. The convergence indicator is presented in Figure 15.
- 4) Contact of a fluid and two anisotropic solids, with IF at the top, AS1 at the bottom left, and AS2 at the bottom right. The convergence indicator is presented in Figure 16.

The numerical results for the corner problem were similar to those obtained for the horizontal interface:

- 1) The contact of three solids produces a second order of convergence for the three FDS.

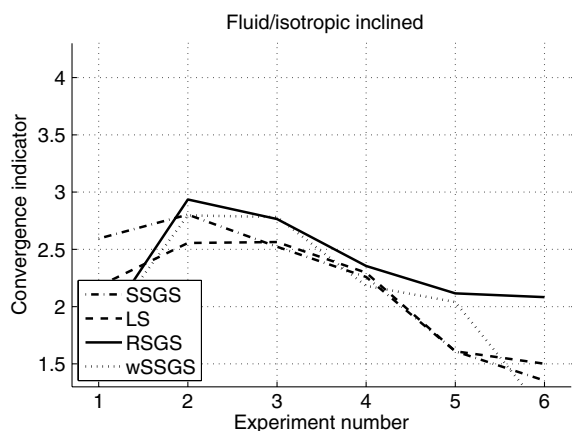


Figure 9. Convergence indicator for the SSGS (dashed-dotted), LS (dashed), RSGS (solid), and SSGS with no parameter modification (dotted) for the model with an inclined fluid-solid interface, isotropic media. The horizontal axis represents the experiment number, and the vertical axis corresponds to the convergence indicator, which is 4 for the second order and 2 for the first order of convergence.

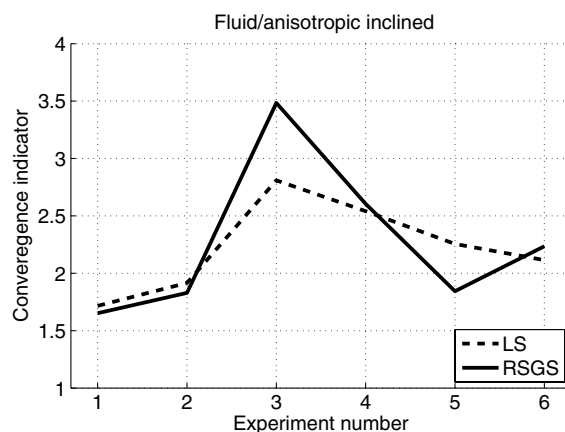


Figure 11. Convergence indicator for the LS (dashed), RSGS (solid) for the model with a horizontal fluid-solid interface, anisotropic media. The horizontal axis represents the experiment number, and the vertical axis corresponds to the convergence indicator, which is 4 for the second order and 2 for the first order of convergence.

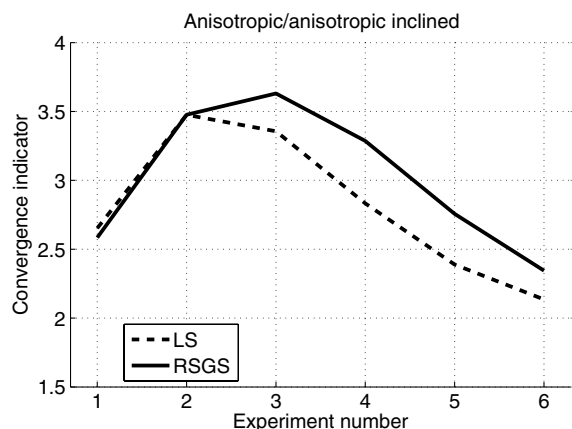


Figure 10. Convergence indicator for the LS (dashed) and RSGS (solid) for the model with a horizontal solid-solid interface, anisotropic media. The horizontal axis represents the experiment number, and the vertical axis corresponds to the convergence indicator, which is 4 for the second order and 2 for the first order of convergence.

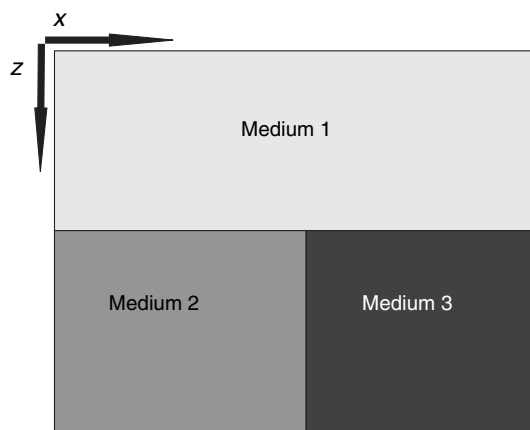


Figure 12. A typical model for the corner problem.

- 2) The existence of a fluid-solid interface reduces the convergence for the LS and RSGS to the first order, whereas the SSGS remains a second order of convergence.

DISCUSSION

Based on the numerical simulations, we experimentally studied the rate of convergence of the three commonly used FDS in elastic media with horizontal/vertical and inclined planar interfaces and corner geometry problems. The main objective of this study was to analyze the accuracy of each of these FDS for modeling seismic-wave propagation in media with interfaces. Therefore, three different approaches were applied:

- 1) the balance technique (Samarskii, 2001);
- 2) averaging of finely layered models (Schoenberg and Muir, 1989; Moczo et al., 2002); and
- 3) reflectivity analysis (Lisitsa et al., 2010; Podgornova et al., 2010).

Our results prove that in the case of planar horizontal/vertical interfaces between two solids, all the schemes can be modified to produce the second-order accuracy by arithmetic averaging of density and finely layered media averaging for the stiffness tensor in the vicinity of the interface, which is consistent with the results of (Moczo et al., 2002). However, for fluid-solid interfaces, a second-order convergence is produced only for the interface between the fluid and isotropic elastic medium when this interface is aligned

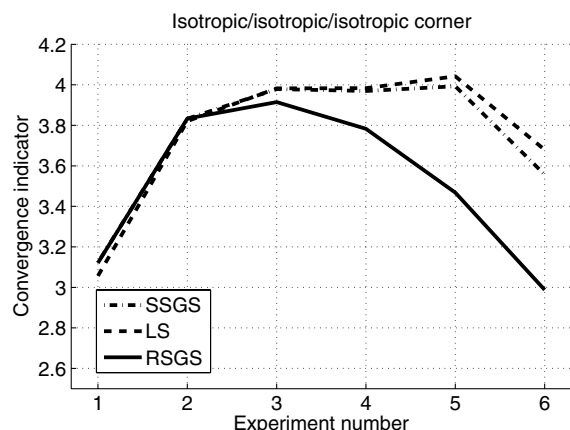


Figure 13. Convergence indicator for the SSGS (dashed-dotted), LS (dashed), RSGS (solid) for the corner problem in which three isotropic solids contact. The horizontal axis represents the experiment number, and the vertical axis corresponds to the convergence indicator, which is 4 for the second order and 2 for the first order of convergence.

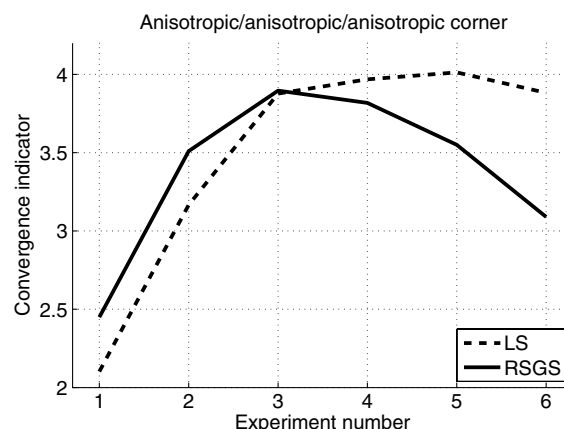


Figure 15. Convergence indicator for the LS (dashed) and RSGS (solid) for the corner problem in which three anisotropic solids contact. The horizontal axis represents the experiment number, vertical axis corresponds to the convergence indicator, which is 4 for the second order and 2 for the first order of convergence.

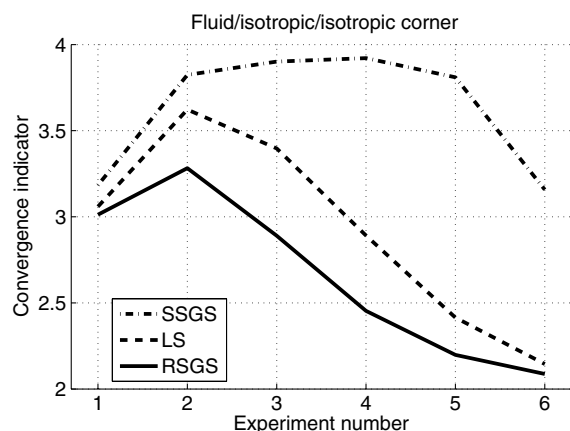


Figure 14. Convergence indicator for the SSGS (dashed-dotted), LS (dashed), RSGS (solid) for the corner problem in which the fluid-filled half-space and two isotropic solids contact. The horizontal axis represents the experiment number, and the vertical axis corresponds to the convergence indicator, which is 4 for the second order and 2 for the first order of convergence.

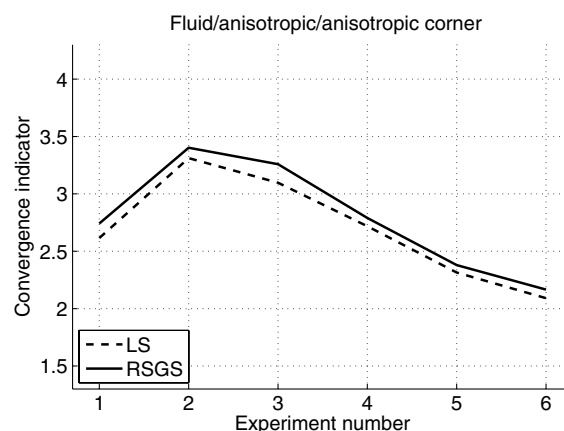


Figure 16. Convergence indicator for the LS (dashed) and RSGS (solid) for the corner problem in which the fluid-filled half-space and two isotropic solids contact. The horizontal axis represents the experiment number, and the vertical axis corresponds to the convergence indicator, which is 4 for the second order and 2 for the first order of convergence.

to the grid line and the SSGS is used with $c_{33} = 0$ ($\mu = 0$ for an isotropic model) at the interface. In addition, neither the LS nor the RSGS preserves a second order of convergence for this geometry.

In addition to the results presented for horizontal/vertical interfaces, a numerical study of two typical situations was performed. The first corresponds to situations when the model is known and must be mapped onto a grid. The simplest staircase approximation was considered, in which the approximation is refined as the grid step becomes smaller. According to the numerical experiments, all three schemes possess approximately the same properties and the convergence rate degenerates to the first order.

Along with these three geometries, the corner problem in which there are three adjacent interfaces was studied. The numerical re-

sults prove that a second-order convergence can be produced for all of the three schemes when there is a contact of three solids if the model parameters are properly modified. However, if a fluid is present, then only the SSGS demonstrates a second-order convergence of the interface error and the LS and RSGS degenerate to a first order.

Therefore, use of the arithmetic mean for density and finely layered averaging (Schoenberg and Muir, 1989) for the stiffness tensor ensures a second order of interface errors convergence when using the SSGS, RSGS, and LS. However, it is only valid for the contacts of solids; for a fluid-solid contact, a second-order convergence is achieved only for the SSGS if the interface is aligned to the grid line and the value of the shear modulus is considered equal to zero at the interface.

Moreover, the effect of the accuracy loss in the presence of the fluid-solid interface can be clearly observed in the snapshots, which are presented in Figure 17. The improper treatment of the fluid-solid interface by the SSGS leads to the appearance of artificial waves that propagate along the interface and the destruction of the wavefront in the vicinity of the interface, even for small grid steps.

CONCLUSION

In this paper a numerical study of the interface error of seismic wave simulation for the three commonly used staggered grid schemes was performed. It was shown that if a planar horizontal/vertical interface between two solids is considered, all three schemes (SSGS, LS, and RSGS) can be modified to produce the second-order accuracy by arithmetic averaging of density and finely layered media averaging for the stiffness tensor in the vicinity of the interface. Same is valid if the model resolution is kept, but the grid used for simulation and wavefield approximation is refined. In this case all the interfaces are determined by a fixed stair-step approximation thus can be treated as horizontal/vertical aligned to the grid lines, thus a second order of convergence can be preserved. However, if an original model contained inclined interfaces and they are reapproximated as the grid is refined, the convergence of the finite-difference solution degenerates to the first order.

In addition, the case of fluid-solid interfaces was considered. It was shown that the only case where the second order of numerical convergence is preserved is the interface aligned to the grid line and the solution is computed by the SSGS, whereas neither the LS nor the RSGS preserves a second-order of convergence for this geometry. If an interface is not aligned to the grid lines or if it is inclined, none of the considered schemes preserve a second order of convergence.

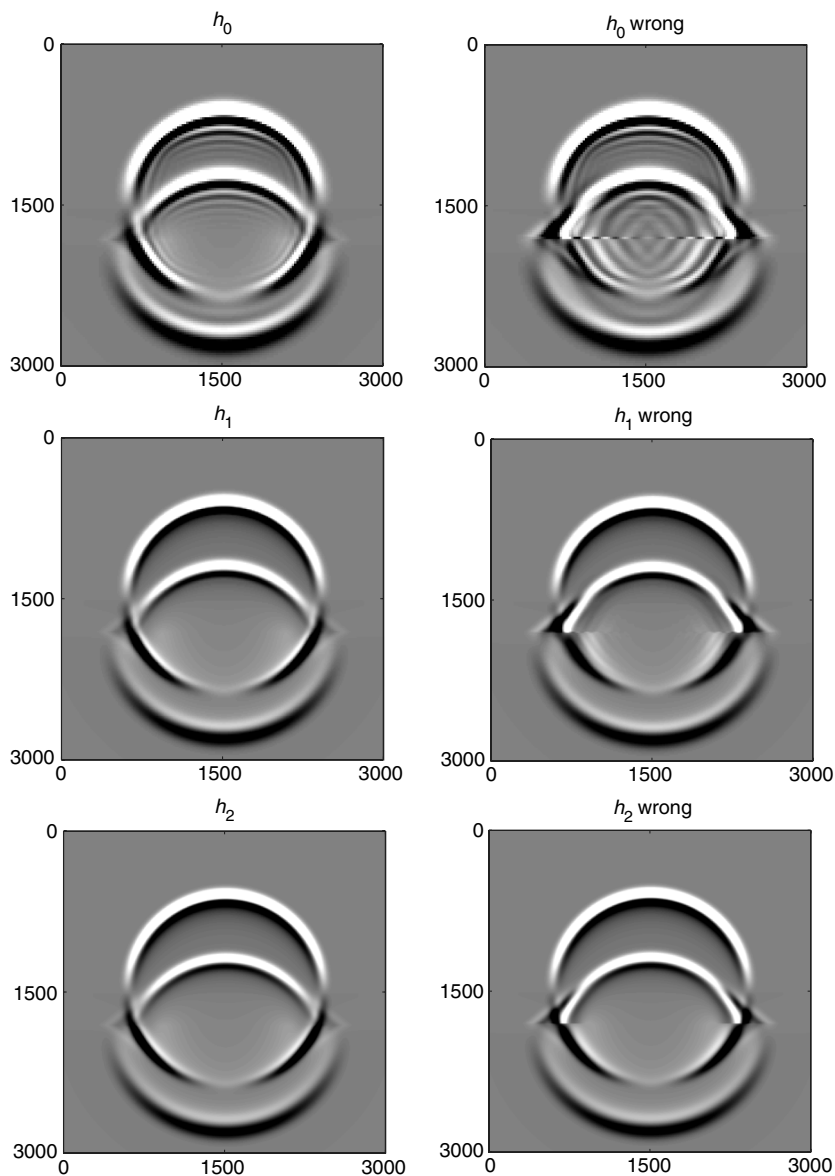


Figure 17. Snapshots of the wavefield computed by the SSGS for the model with a horizontal fluid-solid interface. Axes are in meters, the left column corresponds to the proper modification of the coefficients, the right column corresponds to no modification, and grid steps decrease by a factor of 2 downward.

ACKNOWLEDGMENTS

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APPENDIX A

MODIFICATION OF THE MODEL PARAMETERS

Briefly described here are the methods used for modifying the coefficients in the vicinity of the interfaces for the three FDS (following Gustafsson and Wahlund, 2004; Lisitsa and Vishnevskiy, 2010; Podgornova et al., 2010).

Balance technique

The balance technique as applied to FDs is discussed in detail in Samarskii (2001). This approach establishes a link between FDs and FVs (Brossier et al., 2008). To construct a scheme based on the balance technique, the system of equations should be rewritten in a divergence-free form. To do this, Hooke's law must be resolved with respect to strains and the resulting equations should be integrated over the elementary volume.

To illustrate the application of the balance technique to the elastic-wave equation, consider the first equation of 1. Integrating over a rectangle volume and using the divergence theorem produces the following:

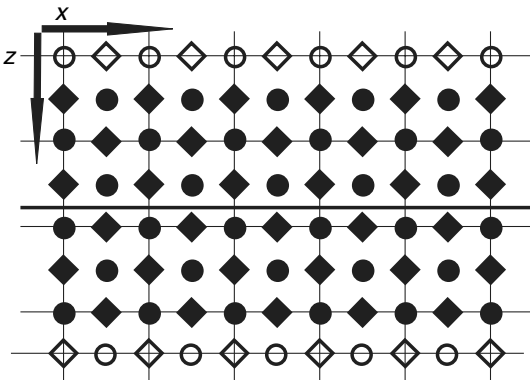


Figure A-1. A grid for the LS with a four-point stencil in the vicinity of the interface when a reflectivity analysis is applied. The interface is represented by a solid line, the points where the model parameters are known are not filled with color, and points where the coefficients are modified based on the reflectivity analysis are filled with color.

$$\begin{aligned} & \int_{x_{I-1/2}}^{x_{I+1/2}} \int_{z_{J-1/2}}^{z_{J+1/2}} \rho(x, z) \frac{\partial u_x(x, z; t)}{\partial t} dx dz \\ &= \int_{z_{J-1/2}}^{z_{J+1/2}} (\sigma_{xx}(x_{I+1/2}, z; t) - \sigma_{xx}(x_{I-1/2}, z; t)) dz \\ &+ \int_{x_{I-1/2}}^{x_{I+1/2}} (\sigma_{xz}(x, z_{J+1/2}; t) - \sigma_{xz}(x, z_{J-1/2}; t)) dx. \end{aligned} \quad (A-1)$$

The function u_x is continuous inside any volume within a medium (see equation 3), so the mean value theorem can be applied to produce

$$\begin{aligned} & \int_{x_{I-1/2}}^{x_{I+1/2}} \int_{z_{J-1/2}}^{z_{J+1/2}} \rho(x, z) \frac{\partial u_x(x, z; t)}{\partial t} dx dz \\ &= \partial_t u_x(x_I, z_J; t) \int_{x_{I-1/2}}^{x_{I+1/2}} \int_{z_{J-1/2}}^{z_{J+1/2}} \rho(x, z) dx dz + O(h^\beta) \\ &= \hat{\rho}_{I,J} \frac{\partial u_x(x_I, z_J; t)}{\partial t} h_x h_z + O(h^\beta), \end{aligned} \quad (A-2)$$

where $\beta = 4$ if the density is continuous inside the volume and $\beta = 3$ if the density is discontinuous. The modified parameter $\hat{\rho}_{I,J}$ is the arithmetic averaging of the density over the volume. The parameter h denotes a grid step under the assumption of a linear relation between the spatial grid steps.

To develop an approximation of the right side of equation A-1 and Hooke's law, we must specify which FDS is used.

Standard staggered-grid scheme

We start with the SSGS. Assume that the interfaces are parallel to the coordinate lines and coincide with the thin black lines in Figure 1. The model parameters and the diagonal components of the stress tensor are to points with integer indices (i, j) (circles in Figure 1), off-diagonal components correspond to half-integer points $(i + 1/2, j + 1/2)$ (crosses in Figure 1), and velocities correspond to points with mixed coordinates: horizontal in $(i + 1/2, j)$ and vertical in $(i, j + 1/2)$. Let us take $I = i + 1/2$ and $J = j$. Both σ_{zz} and σ_{xz} are now continuous in the right side of equation A-1, so the mean value theorem is applicable with the same accuracy as for the left side, and the following FD equation is valid:

$$\begin{aligned} & \hat{Q}_{i+1/2,j} \frac{\partial u_x(x_{i+1/2}, z_j; t)}{\partial t} \\ &= \frac{\sigma_{xx}(x_{i+1}, z_j; t) - \sigma_{xx}(x_i, z_j; t)}{h_x} \\ &+ \frac{\sigma_{xz}(x_{i+1/2}, z_{j+1/2}; t) - \sigma_{xz}(x_{i+1/2}, z_{j-1/2}; t)}{h_z} + O(h); \end{aligned} \quad (A-3)$$

$$\hat{q}_{i+1/2,j} = \frac{q_{i+1,j} + q_{i,j}}{2}. \quad (\text{A-4})$$

The same technique applied for the second equation of 1 produces

$$\begin{aligned} \hat{q}_{i,j+1/2} &= \frac{\partial u_z(x_i, z_{j+1/2}; t)}{\partial t} \\ &= \frac{\sigma_{xz}(x_{i+1/2}, z_{j+1/2}; t) - \sigma_{xz}(x_{i-1/2}, z_{j+1/2}; t)}{h_x} \\ &+ \frac{\sigma_{zz}(x_i, z_{j+1}; t) - \sigma_{zz}(x_i, z_j; t)}{h_z} + O(h); \end{aligned} \quad (\text{A-5})$$

$$\hat{q}_{i,j+1/2} = \frac{q_{i,j+1} + q_{i,j}}{2}. \quad (\text{A-6})$$

The FD expressions A-3–A-6 provide spatial approximations of equations 1 in the vicinity of vertical/horizontal interfaces for two isotropic elastic media.

An analysis of how Hooke's law (equation 2) is approximated by the balance techniques shows that σ_{xx} and σ_{zz} integration should be performed over rectangles with centers at (x_i, z_j) so that coefficients C_{11} , C_{22} , and C_{12} (SSGS is applied for elastic media with $C_{13} = C_{23} = 0$) do not need to be modified. The FD equation for the off-diagonal component of the stress tensor is derived by the integration of the modified differential equations

$$\frac{1}{C_{33}} \frac{\partial \sigma_{xz}}{\partial t} = \frac{\partial u_z}{\partial x} + \frac{\partial u_x}{\partial z}, \quad (\text{A-7})$$

over rectangles centered at $(x_{i+1/2}, z_{j+1/2})$. The application of the mean value theorem produces the following FD relations:

$$\begin{aligned} &\frac{\partial \sigma_{xz}(x_{i+1/2}, z_{j+1/2}; t)}{\partial t} \\ &= (\hat{C}_{33})_{i+1/2,j+1/2} \left(\frac{u_z(x_{i+1}, z_{j+1/2}; t) - u_z(x_i, z_{j+1/2}; t)}{h_x} \right. \\ &\quad \left. + \frac{u_x(x_{i+1/2}, z_{j+1}; t) - u_x(x_{i+1/2}, z_j; t)}{h_z} \right) + O(h), \end{aligned} \quad (\text{A-8})$$

with

$$\begin{aligned} &(\hat{C}_{33})_{i+1/2,j+1/2} \\ &= \left(\frac{1}{(C_{33})_{i,j}} + \frac{1}{(C_{33})_{i+1,j}} + \frac{1}{(C_{33})_{i,j+1}} + \frac{1}{(C_{33})_{i+1,j+1}} \right)^{-1}. \end{aligned} \quad (\text{A-9})$$

The fluid-solid interface should be studied separately for the ideal fluid $c_{33} = 0$, for which formula A-9 is not applicable. However, a commonly used extension of formula A-9 is based on the consideration of a limiting case when some of the summands in formula A-9 tend to zero, which produces the fluid-solid interface for the equality

$$(\hat{C}_{33})_{i+1/2,j+1/2} = 0. \quad (\text{A-10})$$

We prove numerically that assumption A-10 preserves the second order of convergence of the solid-fluid interface error.

Lebedev scheme

If an interface is aligned to the grid lines, then a modification of the mass density for the LS is the same as for the SSGS. As for the stiffness tensor C , two situations should be analyzed. The first corresponds to when C is computed at the center of a grid cell $\hat{C}_{i,j}$, so modification is not required. The second claims to compute $\hat{C}_{i+1/2,j+1/2}$, for which the balance technique is not applicable because either σ_{xx} or σ_{zz} is discontinuous at the interface, which contradicts formula A-1 where the wavefield is supposed to be sufficiently smooth.

If a fluid-solid interface is considered, then the density modification is not clear because the velocity components defined at the interface may also be discontinuous.

Therefore, the balance technique is not applicable for Lebedev FDS.

Rotated staggered-grid scheme

If an interface is aligned to the grid lines and passes through the points marked as rectangles in Figure 3, then only the density at the corners of the cells must be modified as a result of geometry of the grid for the RSGS. The balance technique now produces

$$\hat{\rho}_{i+1/2,j+1/2} = \frac{1}{4}(\rho_{i,j} + \rho_{i+1,j} + \rho_{i,j+1} + \rho_{i+1,j+1}). \quad (\text{A-11})$$

If a fluid-solid interface is considered, the RSGS faces the same problem as the LS. Tangential velocity components may be discontinuous at the interface, which leads to the inapplicability of the balance technique.

Averaging of a finely layered model

As was shown above, the balance technique is a powerful tool for constructing a modification of the FDS coefficients; however, its applicability is restricted by relatively simple cases. The assumption of continuity of the wavefield components inside a cell is not always true, especially for anisotropic elastic-wave equations. To overcome this restriction, Muir et al. (1992) and Moczo et al. (2002) offer a technique similar to the averaging of finely layered media (Backus, 1962; Schoenberg and Muir, 1989). A brief description of the main ideas of this approach follows.

Consider the stress-strain relation 2 for a medium with a horizontal interface. For this situation, σ_{zz} , σ_{xz} , and ε_{xx} are continuous at the interface whereas all the remainder can be discontinuous. Let us reorder the variables and rewrite equation 2 as follows:

$$\begin{aligned} \begin{pmatrix} \sigma^N \\ \sigma^T \end{pmatrix} &= \begin{pmatrix} \sigma_{zz} \\ \sigma_{xz} \\ \sigma_{xx} \end{pmatrix} = \begin{pmatrix} c_{22} & c_{23} & c_{12} \\ c_{23} & c_{33} & c_{13} \\ c_{12} & c_{13} & c_{11} \end{pmatrix} \begin{pmatrix} \varepsilon_{zz} \\ \varepsilon_{xz} \\ \varepsilon_{xx} \end{pmatrix} \\ &= \begin{pmatrix} C^{NN} & C^{NT} \\ C^{TN} & C^{TT} \end{pmatrix} \begin{pmatrix} \varepsilon^N \\ \varepsilon^T \end{pmatrix}. \end{aligned} \quad (\text{A-12})$$

Equation A-12 is resolved with respect to the discontinuous components of the wavefield σ^T and ε^N , and the integral averaging over the cell is applied as follows:

$$\begin{aligned}
\langle \sigma^T \rangle &= \langle C^{TT} - C^{TN}(C^{NN})^{-1}C^{NT} \rangle \varepsilon_{I,J}^T \\
&\quad + \langle C^{TN}(C^{NN})^{-1} \rangle \sigma_{I,J}^N, \\
\langle \varepsilon^N \rangle &= - \langle (C^{NN})^{-1}C^{NT} \rangle \varepsilon_{I,J}^T + \langle (C^{NN})^{-1} \rangle \sigma_{I,J}^N,
\end{aligned} \tag{A-13}$$

where $\langle \rangle$ stands for the integral average over the cell and $\sigma_{I,J}^N$ and $\varepsilon_{I,J}^T$ are the values of the corresponding functions at the center of the cell.

This technique allows for the modification of coefficients for the following cases:

- 1) interface aligned to a grid line, including the LS,
- 2) interface parallel but not aligned to a grid line for all three schemes, and
- 3) flat inclined interface according to Muir et al. (1992) and smoothly curved interface according to Moczo et al. (2002).

However, this approach is not free from theoretical drawbacks. First, it cannot be applied to the fluid-solid interfaces because of the degeneration of the stiffness tensor C^{NN} for ideal fluids. Second, there is only empirical justification of this approach for inclined interfaces that results in the stiffness tensor of a general form, even when two isotropic solids contact. Additionally, there is no proof of a second-order convergence when this technique is applied.

Reflectivity analysis

The main principles of an approach to coefficient modification based on the straightforward study of the convergence rate of numerical solutions will now be described. To perform such a modification, a simple model is considered where two homogeneous subspaces are separated by a planar interface. This interface generates reflected and transmitted waves for any incident plane wave. The amplitudes of these waves are given by a vector X satisfying the system of linear algebraic equations derived from contact conditions following Aki and Richards (1980):

$$\mathcal{M}X = F, \tag{A-14}$$

Matrix $\mathcal{M}$ in formula A-14 depends on the elastic parameters of the model in the vicinity of the interface and the vector F is the right side depending on the type of incident wave.

A similar problem can be formulated for an FDS. Let all of the coefficients of the scheme be known inside the homogeneous parts of the model up to a certain distance from the interface. The distance depends on the width of the stencil; for a two-point stencil, it can be one-half a step on each side of the interface, and for a four-point stencil $M = 1$ in equations 6 and 7, the distance is $3h/2$, etc. (see Figure A-1 for additional details).

Thus, a system of linear algebraic equations for the reflection/transmission coefficients of the FD problem is constructed (Cohen and Joly, 1996; Gustafsson and Wahlund, 2004; Lisitsa et al., 2010, 2012):

$$\hat{\mathcal{M}}(h, \vec{\alpha}) \hat{X}(h, \vec{\alpha}) = \hat{F}(h, \vec{\alpha}). \tag{A-15}$$

Matrix $\hat{\mathcal{M}}$, solution $\hat{X}$, and right side $\hat{F}$ in equation A-15 are dependent on the grid steps h and the set $\vec{\alpha}$ of unknown coefficients of the FDS.

The FDS parameters that minimize the norm of a difference between these two sets of reflection/refraction coefficients is found by

$$\min_{\vec{\alpha}} \|X - \hat{X}(h, \vec{\alpha})\| = O(h^2). \tag{A-16}$$

As is shown in Podgornova et al. (2010) and Lisitsa et al. (2010) for the second-order schemes, i.e., $M = 0$ in formulas 6 and 7, a set of unknown coefficients can be recovered in a unique way, and these coefficients coincide with formulas for the averaging of a finely layered medium and balance technique. For schemes with a wider stencil, the solution $\vec{\alpha}$ of equation A-16 is not unique, but formulas for the averaging of a finely layered medium preserve the second order of convergence. An attempt to increase the order of convergence of the reflection/transmission coefficients produces frequency-dependent coefficients that are difficult to implement in practice.

The main advantage of the approach based on reflectivity analysis is the possibility of studying a fluid-solid interface. In this case, a second order of convergence can be reached only for the SSGS and only if the interface is aligned with a grid line. In other cases, including the LS and RSGS, a second-order convergence cannot be achieved, despite the geometry of the interface and a method of modifying the model parameters (see Podgornova and Lisitsa, 2010).

APPENDIX B CONVERGENCY INDICATOR

What follows is a brief review of the theory of FDS that are applied to the linear differential problem $L[u] = f$, where L is a differential operator, $u(x)$ and $f(x)$ are from the functional spaces M and D with norms $\|\dots\|_M$ and $\|\dots\|_D$, and argument $x \in \Omega \subset R^N$. The FD approximation of this statement is constructed by sampling X with a step $\vec{h} = \{h_1, \dots, h_N\}$ and an approximation of differential operator L by matrix L_h . The solution of the system of linear algebraic equations $L_h u_h = f_h$ converges to the solution in the differential formulation $\Leftrightarrow \|u_h - [u]_h\|_{M_h} \rightarrow 0$ for $h \rightarrow 0$, where M_h is a suitable finite-dimensional space with an appropriate norm and $[u]_h$ is the projection of u onto M_h .

It is said that the convergence rate is equal to k if the following relation holds:

$$\frac{\|u_h - [u]_h\|_{M_h}}{\|\vec{h}\|^k} = C + c_1(\vec{h}), \tag{B-1}$$

with $c_1(\vec{h}) \rightarrow 0$ for $\vec{h} \rightarrow 0$. This relation is the same as

$$u_h = [u]_h + C e_h h^k + h^k f_h c_1(h), \tag{B-2}$$

where e_h and f_h are unit vectors in M_h .

The straightforward application of equation B-2 for $h_j = h/2^j$ and $h/2$ introduces the convergency indicator (see equation 13):

$$\delta_j = \frac{\|u_{h_j} - P_{h_j} u_{h_{j+1}}\|_{M_{h_j}}}{\|u_{h_{j+1}} - P_{h_{j+1}} u_{h_{j+2}}\|_{M_{h_{j+1}}}} = 2^k + c_2(h), \tag{B-3}$$

where $c_2(h) \rightarrow 0$ for $h \rightarrow 0$. The operators P_{h_j} are projectors from $M_{h_{j+1}}$ to M_{h_j} , which only save the corresponding components of the vector $u_{h_{j+1}}$.

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